Provenance-aware auditing of censored fatigue-life bounds for laser-powder-bed-fused 316L

A public-data benchmark with a failed prespecified selective-decision hypothesis; research draft, 20 October 2026 | Working draft; author names and affiliations not assigned.

Abstract

Uncertainty estimates for literature-trained fatigue models require evaluation at the source and decision levels relevant to their use. We audit lower fatigue-life bounds for laser-powder-bed-fused ordinary 316L using 1,126 public observations, including 139 right-censored runouts, from 120 conditions and 37 publication DOIs. Pre-fit near-identical-curve screening defines 29 conservative provenance blocks, without claiming verified specimen identity. A locally registered three-seed leave-block-out protocol compares censored Basquin-style and richer linear models, endpoint and complete-cohort calibration, and fixed support/instability rejection rules. The richer linear model's raw lower-decile bound has an equal-block empirical violation identification interval of 28.97-30.77%. Endpoint calibration reduces this to 14.50-15.40%; cohort-max calibration gives 0.197% at 3.541 decades of conservatism relative to fitted median life. The latter result changes materially under a weaker-source-family sensitivity. At matched mean availability of 83.39%, the combined rejection rule does not meet its frozen improvement criterion against either component or random ranking; its true-risk contrasts remain unidentified in sign. All primary neural fits reach their numerical cap; a separately labelled longer-iteration sensitivity converges but does not establish architecture ranking. The result is a reproducible negative decision audit, not a reproduction of the original SS316L Bayesian network, a new R90C90 procedure, a guarantee under arbitrary process shift or physical component validation.

1. Introduction

Fatigue design of laser-powder-bed-fused metals must account for processing conditions, defects, surface state, post-processing and substantial experimental scatter. A useful life prediction should therefore describe its uncertainty and its limits of applicability. In particular, an apparently conservative life bound can become optimistic when applied to a publication or manufacturing condition absent from training. We investigate this decision-oriented question using publicly available 316L fatigue records, with provenance and right censoring retained throughout.

Centola et al. compare neural fatigue models and learn median and R90C90 stress-life representations for additively manufactured SS316L [1]. Their subsequent work connects probabilistic predictions to topology constraints [2] and transfers information across alloys under limited target-alloy data [3]. The latter already evaluates withheld experimental datasets. The topology study also checks final risk volume and explicitly discusses physical validation for critical applications. These strengths motivate complementary auditing; they do not justify alleging that the authors ignored grouping, censoring or component-scale limitations.

Related work substantially limits novelty. Wietecha-Reiman et al. model publication-associated variability in historical AM stainless-steel data [5]. Di Maggio et al. withhold complete fatigue datasets when assessing learning algorithms [6]. Wu et al., whose metadata were available at the search cutoff although the issue is dated 2027, compare splitting strategies and physics baselines [7]. Conformal fatigue uncertainty and local undercoverage are also being studied [8]. Hierarchical and survival prediction require explicit assumptions [9,10], while distribution shift can invalidate ordinary exchangeability arguments [11]. Selective prediction predates the present application [12].

Our contribution is consequently an auditable empirical study, rather than a new general uncertainty algorithm. We distinguish specimen-level and complete-cohort targets, quantify censoring-related non-identification, and test a fixed support-plus-instability rejection rule against its components at matched availability. The locally registered primary intervention fails its

positive-claim criterion. We report that negative result together with the provenance and calibration sensitivities that determine its interpretation.

2. Data provenance and benchmark population

The source is FatigueData-AM2022, Figshare version 2 [4,13]. Raw bytes, checksums, units, observation addresses and transformations are retained. Flattening reproduces 15,146 S-N observations from 1,610 datasets and 459 source DOIs. A 100-record MAT-versus-JSON coordinate check agrees; this establishes format consistency, not experimental accuracy. We select exact nominal ordinary-316L labels and L-PBF processing, excluding deliberately modified alloys. The broader subset contains 1,473 observations from 160 conditions and 46 DOIs.

The primary population is further restricted to database-labelled uniaxial, force-controlled tests at 20-30 degrees C and nominal stress-concentration factor 0.99-1.01. Four strain-controlled records labelled S-N were excluded before fitting. The resulting cohort contains 1,126 observations, 120 conditions and 37 DOIs: 987 failures and 139 runouts. No lifetime cutoff, residual-based outlier exclusion or failure-only selection was applied. The descriptor sometimes supplies default temperature, environment, loading and concentration-factor labels [4]; eligibility therefore reflects database metadata rather than independently certified experimental settings.

Source DOI alone does not establish independent specimens. A pre-fit screen required the same loading ratio, mode and censoring status and identified uniquely matched coordinates within 2% in cycles and 1% in stress. Edges with at least five matches covering at least 85% of the smaller curve defined three connected publication families. These families reduce 37 DOIs to 29 conservative evaluation blocks. Near-identical digitized curves support preventing cross-split reuse, but do not prove physical specimen identity. The primary analysis retains all observations; removal is examined separately.

Original-source spotchecks compared six ordinary-316L observations in Stern et al. [14] and thirteen observations in Voloskov et al. [15] with their published figures. All nineteen matched, with no runout disagreements; the largest approximate discrepancy was 2.60% in cycles and 0.25% in stress. Maximum-stress-to-amplitude conversion was checked using sigma_a=(1-R)sigma_max/2. These checks establish agreement with plotted data, not the authors' raw specimen records, which were not obtained. Voloskov's borrowed comparison series were distinguished from its own observations.

Processing information is incomplete. In the primary cohort, power is missing for 548 records, scan speed for 533 and hatch spacing for 684; heat-treatment temperature and time are each missing for 816. Missing or range-valued scalar entries remain uncertain. We neither invent processing parameters nor interpret imputed values as measured physics. Sixteen of Centola's seventeen source DOIs occur in the broader ordinary-316L subset, and twelve occur in the primary subset. This is source overlap, not a verified match to their 518 observations or independent external validation.

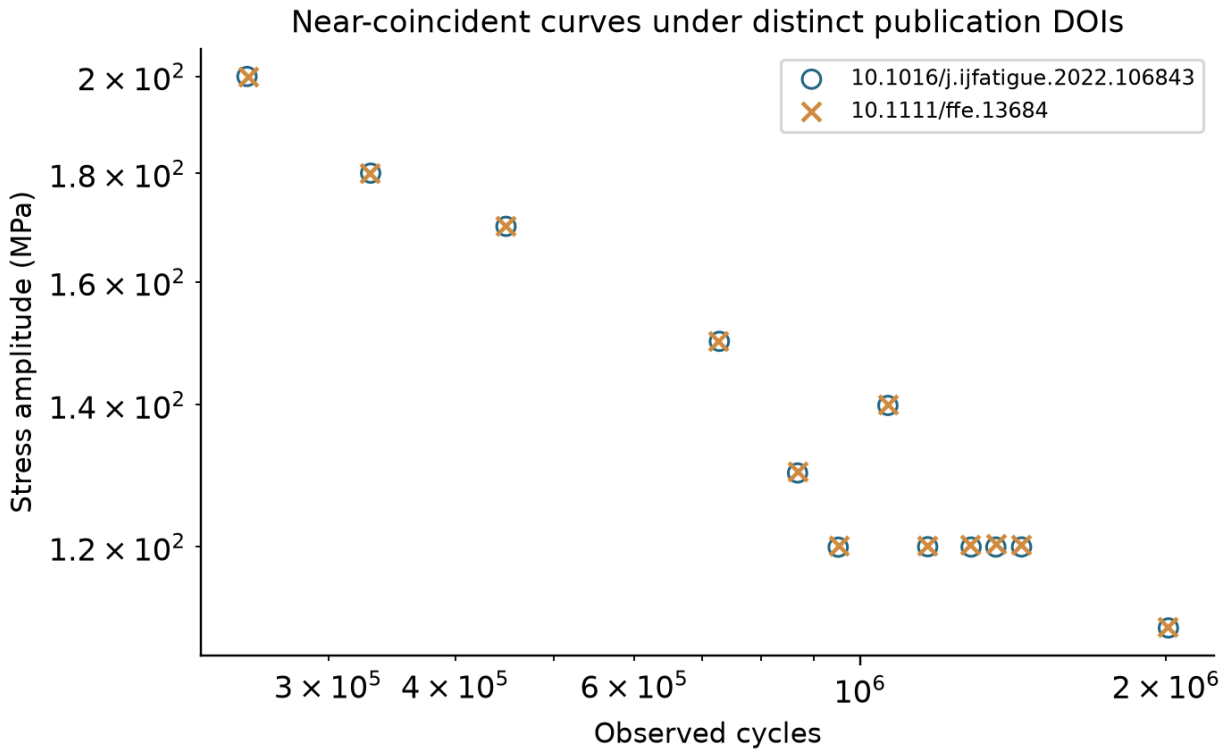


Figure 1. Two source-DOI curves nearly coincide in physical coordinates. This is a conservative provenance-screening signal, not verified specimen identity. Three multi-DOI families are blocked together in primary tests. Data: FatigueData-AM2022 v2 [4,13]; the complete three-family graphic and point matches accompany the code.

3. Prespecified methods

The primary protocol and pre-fit amendments were saved locally before real-data fitting. A cohort-target clarification was recorded during execution before aggregate test inspection; the numerical-extension sensitivity was added after observing a convergence failure and is labelled exploratory. This was a local preregistration, not a public independent registry. The primary descriptive question concerns the reliability-conservatism tradeoff of lower-life bounds. The intervention hypothesis, H2, requires the combined gate to reduce known violation rates against distance, instability and random ranking at the primary matched accepted fraction 0.8 in every seed, with all paired descriptive 95% interval upper endpoints below zero. Favorable secondary analyses cannot rescue a failed criterion.

Each outer fold withholds one complete provenance block. Seeds 0, 17 and 42 generate fitting/calibration partitions of the remaining blocks: nine calibration blocks and nineteen fitting blocks for the primary 29-block analysis. Thus each observation is tested once per seed. Training alone determines numeric medians, scales and categorical encodings, including missing and unseen-category indicators. Neither source identifiers nor post-test runout status enters the prospective feature vector. Test outcomes do not select models, calibration shifts or gate thresholds.

Let Y=log10(min(T,C)), delta=1 for observed failure and delta=0 for runout, and log10(T)|X ~ Normal(mu_theta(X),sigma^2). With z=(Y-mu_theta(X))/sigma, the fitted negative log likelihood is -sum_i w_i{delta_i[log phi(z_i)-log sigma]+(1-delta_i)log[1-Phi(z_i)]}, plus the fixed ridge penalty. Weights give every fitting block equal total likelihood contribution. The lognormal specification assumes a common conditional scatter and appropriate conditional censoring; neither assumption is established merely by fitting the model.

Three fixed models were implemented: a Basquin-style linear location using log stress amplitude and load ratio; a richer linear location additionally using orientation, power, scan speed, hatch spacing, layer thickness, heat-treatment temperature/time and surface/thermal categories; and a twelve-unit tanh location network. Linear and network penalties are 0.001 and 0.01, respectively. All use the same censored likelihood. The primary numerical caps are 1,000 and 1,500 iterations.

These are accessible related models, not reproductions of Centola's variational network, risk-volume inputs, SWT treatment or Owen-derived targets.

The raw lower-decile predictor in log cycles is b(X)=mu_hat(X)+Phi_inverse(0.1)sigma_hat. Endpoint calibration forms residuals r_i=b(X_i)-Y_i on calibration observations and subtracts their order statistic at rank ceil[0.9(n_cal+1)]. Cohort calibration first takes the maximum residual within each calibration block and applies the analogous rank across blocks. If the rank exceeds the number of scores, the cutoff is infinite and the life bound zero. With nine calibration blocks the cohort cutoff is their maximum. No such vacuous bound occurred in the primary analysis.

These calibrations have different targets. Endpoint calibration predicts the observed endpoint, which does not exceed true life; it is a deliberately conservative proxy only when the relevant endpoint calibration argument is valid. It is not the censoring-adjusted survival-conformal method of [9]. Maximum-block calibration targets the event that every member of a complete new cohort is covered, and depends on cohort size and sampling. Row dependence and unknown publication shift prevent importing a nominal guarantee without checking assumptions. Neither construction estimates R90C90, nor guarantees source-conditional or accepted-part reliability.

The support score is minimum root-mean-square Euclidean distance to fitting rows in the fixed encoded representation. Instability is the standard deviation of rich-linear locations across eight source-bootstrap fits. Their combined score averages calibration empirical-CDF ranks. Bootstrap instability is neither Bayesian weight uncertainty nor individual-life scatter. Operational acceptance thresholds are calibration-score quantiles fixed at 0.5 and 0.8. In the separate retrospective comparison, each held-out block accepts ceil(f*n) cases ranked by each score, ensuring identical counts for all methods. Outcome-blind random ranking averages twenty fixed realizations; ties use deterministic row-ID/seed hashes.

Danger is the strict event L>T, where L is the predicted lower life. A failure determines its status. For a runout, T>C: L<=C proves no violation, whereas L>C is unresolved. If d accepted failures are known violations and u accepted runouts are unresolved among n accepted observations, empirical risk is partially identified by [d/n,(d+u)/n]. We also evaluate L>2T. These identification bounds are distinct from confidence intervals. Whole-cohort violation is likewise bounded using whether any member is known or possibly to violate.

Primary summaries average seeds within each provenance block and then weight blocks equally. Pooled-row rates are secondary. Paired 2,000-resample block bootstraps preserve method pairing, but their 95% intervals are descriptive because outer training sets overlap. Three seeds do not create three independent experiments. Conservatism is the block-averaged within-block median of mu_hat-log10(L), measured in decades. We report availability, accepted-block coverage, stress and censoring mix, failure-only error, censored likelihood and numerical diagnostics. Changes in known violations alone cannot establish true-risk improvement when the corresponding identification contrast crosses zero.

4. Primary results

The rich-linear raw lower-decile bound has an equal-block empirical violation interval of 28.97-30.77%, despite its nominal lower-decile interpretation. Its known-violation rate ranges from 0% to 100% across seed-averaged held-out blocks; five blocks have zero known violations and twenty-one exceed 10%. Such heterogeneity concerns this fitted model and corpus. It is not evidence that every Bayesian or process-informed fatigue model is unreliable.

Pooled endpoint calibration lowers the rich-linear interval to 14.50-15.40%, while increasing mean conservatism from 0.619 to 1.134 decades. Cohort-max calibration lowers specimen-level violation to 0.197%, with no residual censoring ambiguity, but requires 3.541 decades of conservatism relative to the fitted median life. The small violation count therefore cannot be evaluated without its substantial loss of useful life. It is not an observed 90% reliability-and-confidence design certificate.

Evaluating the cohort method against its own target gives four known failed cohort-seed instances among 87, or a seed-averaged block rate of 4.60%. Endpoint-calibrated linear bounds fail

simultaneously in at least 52.87% and possibly 55.17% of cohort-seed instances. These are descriptions of complete retained cohorts, not 87 independent experimental campaigns. The cohort-size target explains why small average specimen risk and simultaneous-cohort performance must be reported separately.

At the primary matched fraction 0.8, integer ceiling produces mean accepted availability 83.39%, rather than exactly 80%. The combined gate gives a rich-linear endpoint-bound risk interval of 14.23-14.89%; distance-only gives 14.10-14.57%, instability-only 14.29-15.13%, and random ranking 14.60-15.51%. Every block retains observations. The combined-minus-distance known-risk difference is +0.130 percentage points, with a descriptive 95% interval [-0.639,+0.891]; versus instability it is -0.063 [-1.342,+0.909], and versus random -0.377 [-1.165,+0.273].

H2 fails the frozen positive-claim rule. Distance comparisons change sign across seeds, instability comparisons also change sign, and all three descriptive intervals include zero. All corresponding all-observation identification contrasts include zero. Consequently, a reliable true-risk improvement is not identified. At the secondary matched fraction 0.5, availability is 50.76%; favorable point differences remain insufficient to overturn the primary result. Operational combined thresholds achieve different availability, including 85.77% at the nominal 0.8 threshold, illustrating why calibrated score quantiles do not promise a fixed future acceptance fraction.

The Basquin-style model provides a serious benchmark: its uncalibrated equal-block violation interval is 14.04-14.97%, and its failure-only mean absolute log-life error is 0.531 decades versus 0.750 for rich linear. Equal-block censored negative log likelihood is 1.128 versus 2.288. These prespecified descriptive comparisons give no evidence that adding the available processing metadata automatically helps. All Basquin, rich-linear and 696 bootstrap fits converge; no fitted scatter reaches its bounds. All 87 primary network fits reach the 1,500-iteration cap, so network scores cannot support an architecture ranking.

Model / bound	Specimen violation (%)	Cohort violation (%)	Cost (decades)
basquin / q10	14.04-14.97	57.47	0.837
basquin / endpoint	12.14-12.85	57.47	0.877
basquin / cohort max	0.04	4.60	2.660
linear / q10	28.97-30.77	71.26	0.619
linear / endpoint	14.50-15.40	52.87-55.17	1.134
linear / cohort max	0.20	4.60	3.541

Table 1. Equal-provenance-block summaries, averaging seeds within blocks. Ranges identify unresolved censoring, not sampling confidence. Cost is the mean within-block median of log10(model median / lower bound). Different bounds target different prediction events.

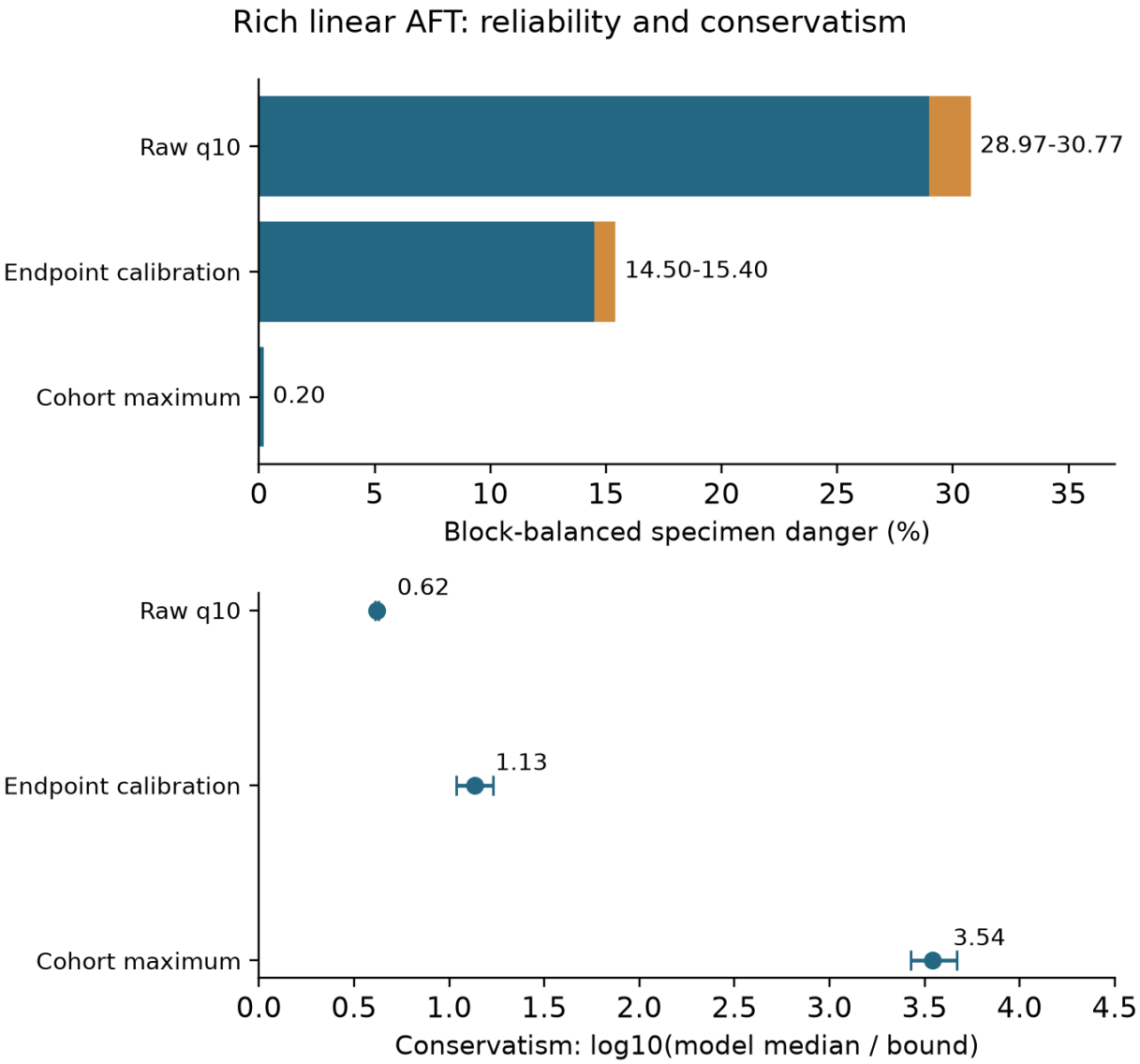


Figure 2. Rich-linear AFT risk and conservatism. Orange segments represent unresolved runouts; lower-panel whiskers are descriptive block-bootstrap 95% intervals. The cohort-maximum target concerns simultaneous whole-cohort coverage. The small primary cohort risk is sensitive to grouping/calibration partitions.

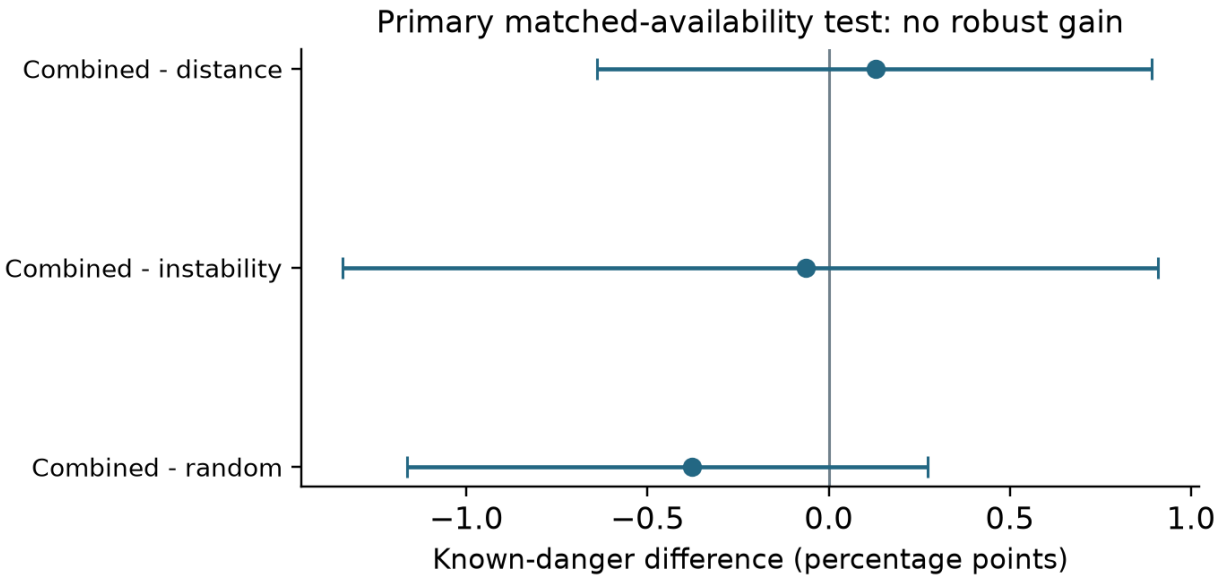


Figure 3. Primary matched-availability contrasts: combined minus each comparator in known violations, with descriptive paired block-bootstrap 95% intervals. The target is 80% per block; integer rounding yields 83.39% mean availability. All intervals cross zero and the frozen superiority rule fails.

5. Sensitivity, numerical reproducibility and audit

Removing 184 near-coincident cross-publication rows under the frozen deterministic rule leaves 942 observations. The rich-linear raw bound then gives 28.91-30.71% violation and the endpoint bound 15.00-15.72%, close to the primary interpretation. Removing near matches may discard distinct replicates, so this sensitivity does not replace the primary provenance-preserving dataset.

Merging one weaker three-of-four-point source-pair candidate produces 28 blocks. Raw and endpoint intervals become 28.49-30.41% and 16.03-16.88%. More strikingly, cohort-max specimen risk increases to 3.67-3.94%, conservatism falls to 2.584 decades, and simultaneous-cohort violation reaches 17.86%. Thus the favorable primary cohort result is not robust to this grouping/calibration perturbation. Coarse calibration, changing fit/calibration composition and cohort structure preclude attributing that difference to one isolated mechanism.

The DOI-only sensitivity permits related curves to cross folds and changes the evaluation population. Reaggregation on the same 29 primary blocks still gives a lower raw rich-linear violation interval, 21.21-22.44%, than primary blocking. Random-row diagnostics give failure-only error 0.393 decades and raw violation 11.04-12.08%. Neither comparison isolates leakage causally: training composition, calibration and the dependence structure also change. Their purpose is to expose how the split changes the apparent task.

Naively treating training runout endpoints as failures yields a raw interval of 27.51-29.09% and an endpoint-calibrated interval of 15.57-16.22%. Apparently improved failure-only fit would not validate this misspecified likelihood. The valid evaluation retains the original censoring statuses. A separate code and arithmetic audit checks split disjointness, training-only encodings, all directional-risk counts and primary contrasts, and replays calibration and bootstrap gates on named folds. A deterministic fold rerun and retained checkpoints support computational reproducibility. These checks are not an independent scientific replication or newly collected validation data.

A post-result numerical investigation increased only the neural iteration cap from 1,500 to 12,000, retaining all other settings and every original fold/seed. All 87 extended fits converged by the original stopping criterion, at no more than 9,183 iterations. The extended model has violation identification intervals of 40.17-44.93% for raw q10, 9.30-10.06% for pooled endpoint calibration and 0.234-0.246% for cohort-max calibration. These are separately labelled numerical-sensitivity results, not replacements for preregistered results or a best-of-settings choice. They do not establish that neural models in general, or the original BNN, perform better or worse.

For the primary rich-linear cohort-max bound, 4.23% of pooled prediction instances yield a lower recommendation below one cycle; its pooled median lower bound is about 141 cycles. These are output-magnitude diagnostics over repeated seed predictions, not independent specimen counts. They illustrate why an extremely low empirical violation rate need not constitute a useful high-cycle fatigue recommendation. No design-life requirement was optimized after inspecting test outcomes.

6. Interpretation and limits

The central empirical result is a tradeoff, accompanied by a failed intervention. Raw model quantiles can be optimistic for unseen literature blocks; calibration can reduce observed violations, but its target and usefulness matter. Extremely short lower-life bounds can achieve small retrospective violation fractions without being useful for design. The tested combination of encoded support and eight-bootstrap instability did not reliably outperform either component at matched availability. This rejects the strong claim for this frozen rule, not all possible support-aware policies.

Relevance to process-to-life and topology tools is practical but bounded. A prediction system could report source support, a censoring-identifiable risk summary and the cost of conservative calibration before its output enters a design constraint. The current study cannot prescribe an allowable stress for a new part. It contains no topology optimization, finite-element mesh study, manufactured component or physical fatigue campaign. Centola's raw data and code were not obtained for this study; the paper reports data available on request. Reproducing the 518-row result would require specimen labels, source lineage, preprocessing, fitted-curve targets and split/code details.

Several limitations constrain extrapolation. Provenance blocks identify near-coincident curves rather than certified manufacturing batches; undiscovered reuse may remain. Experimental literature is selectively published and not sampled from a defined future-process population. Missing processing variables and database defaults can confound source differences. Distances depend on encoding, category proliferation and the unit-scale convention for constant or all-missing features. Homoscedastic lognormal scatter can miss changing failure mechanisms. Censoring may be informative, and unresolved runouts prevent point identification of some risks. Neither block bootstrap intervals nor finite-sample quantile formulas repair these limitations.

The next discriminating validation would use genuinely new source-labelled specimens, with measured process settings, predeclared censoring thresholds and a fixed prospective acceptance rule. A small independent campaign spanning a supported condition and a deliberately shifted condition would be more informative than selecting another method on this test corpus. Before submission, the targeted literature search should be refreshed and inaccessible full texts and supplements checked. Current evidence supports a reproducible negative audit and methodological caution; it does not establish a new universal reliability method.

Authorship, affiliation, individual contributions and any institutional or journal disclosures remain to be supplied by the actual contributors. The code and draft were prepared in an agent-assisted local research workflow. This working draft is suitable for technical review; it is not a claim of peer review, accepted publication or externally validated design authority.

Empirical identification intervals account only for right-censoring ambiguity. They do not propagate digitization error, uncertain database defaults, undiscovered reuse or model/input error. Approximate coordinates can change strict violation labels near a boundary; the19 source spotchecks do not establish a global error bound.

7. Conclusion

On 1,126 public ordinary-316L observations grouped into 29 conservative provenance blocks, a richer censored linear model's lower-decile bound violates observed fatigue life much more often than its nominal interpretation suggests. Endpoint and complete-cohort calibration alter that behavior at different costs, with cohort results sensitive to grouping. The prespecified combined support-instability gate does not satisfy its improvement criterion. The defensible deliverable is the traceable benchmark, explicit distinction between censoring identification and sampling uncertainty, and a negative selective-decision result. Its value lies in making the reliability question testable without overstating reproduction, novelty or physical validation.

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